# LAB 5 File Handling & Exceptions

```
file = open("ABC.txt", "w")
file.write("Hello World\n")
file.write("This is a text file\n")
file.write("It contains some text\n")
file.close()
```

**Assignment:**

1. Write a function in python to read the content from a text file "ABC.txt" line by line and display the same on screen.

```
file = open("ABC.txt", "r")
lines = file.readlines()
for s in lines:
  print(s)
file.close()
```

```
Hello World

This is a text file

It contains some text
```

2. Write a function in Python to count and display the total number of words in a text file “ABC.txt”

```
file = open("ABC.txt", "r")
str = file.read()
words = str.split()
print("Total number of words:", len(words))
file.close()
```

```
Total number of words: 11
```

3. Write a function in Python to count uppercase character in a text file “ABC.txt”

```
file = open("ABC.txt", "r")
str = file.read()
count = 0
for char in str:
  if char.isupper():
    count += 1
print("Total number of uppercase characters:", count)
file.close()
```

```
Total number of uppercase characters: 4
```

4. Write a function display_words() in python to read lines from a text file "story.txt", and display those words, which are less than 4 characters.

```
file = open("story.txt", "w")
file.write("Python is a high-level, general-purpose programming language known")
file.write("for its simple, readable syntax that often uses English-like ")
file.write("keywords, making it an excellent choice for beginners. Created by ")
file.write("Guido van Rossum in 1991, the language emphasizes code clarity and ")
file.write("uses indentation to define code blocks, unlike many other ")
file.write("languages that use semicolons or curly braces.")
file.close()
file1 = open("story.txt", "r")
words = file1.read().split()
for word in words:
  if len(word) < 4:
    print(word)
file1.close()
```

```
is
a
its
it
an
for
by
```

```
van
in
the
and
to
use
or
```

## Assignment:

1. Write a Python program to handle a ZeroDivisionError exception when dividing a number by zero.

```
try:
  num = int(input("Enter a numerator: "))
  den = int(input("Enter a denominator: "))
  result = num / den
except ZeroDivisionError:
  print("Error: Division by zero")
else:
  print("Result:", result)
```

```
Enter a numerator: 10
Enter a denominator: 0
Error: Division by zero
```

2. Write a Python program that prompts the user to input an integer and raises a ValueError exception if the input is not a valid integer.

```
try:
  num = int(input("Enter an integer: "))
except ValueError:
  print("Error: Invalid input")
else:
  print("Integer:", num, "  valid")
```

```
Enter an integer: r
Error: Invalid input
```

3. Write a Python program that opens a file and handles a FileNotFoundError exception if the file does not exist.

```
try:
  file = open("file.txt", "r")
except FileNotFoundError:
  print("Error: File not found")
else:
  print("File opened successfully")
  file.close()
```

```
Error: File not found
```

4. Write a Python program that prompts the user to input two numbers and raises a TypeError exception if the inputs are not numerical

```
try:
  num1 = input("Enter first number: ")
  num2 = input("Enter second number: ")
  float(num1)
  float(num2)
except TypeError:
  print("Error: Invalid input")
else:
  print("Numbers are valid")
```

```
Enter first number: 55
Enter second number: 77
Numbers are valid
```